MATH261 Homework Set 7

1. $\nabla \cdot \mathbf{F} = 4y; \nabla \times \mathbf{F} = \mathbf{0}$; yes
2. $\nabla \cdot \mathbf{F} = 2x\cos(x^2) + 2ye^{y^2}; \nabla \times \mathbf{F} = 1\mathbf{k}$; No
3. $\nabla \cdot \mathbf{F} = 2x; \nabla \times \mathbf{F} = (x - 2z)\mathbf{i} - y\mathbf{j}$; No
4. $\nabla \cdot \mathbf{F} = 0; \nabla \times \mathbf{F} = (-3x^2)\mathbf{j}$; No
5. $\nabla \cdot \mathbf{F} = 0; \nabla \times \mathbf{F} = \mathbf{0}$; Yes
6. $\nabla \cdot \mathbf{F} = -xy\cos(y); \nabla \times \mathbf{F} = \mathbf{0}$; Yes
7. $\nabla \cdot \mathbf{F} = 7; \nabla \times \mathbf{F} = 2\mathbf{i}$.
8. $\nabla \cdot \mathbf{F} = -9; \nabla \times \mathbf{F} = 3\mathbf{j} + 4\mathbf{k}$.
9. $f = x^2 + 3y + 2z^2$
10. $f = xy^2 + ye^z$
11. $f = x\cos(z) + y^2$
12. 16
13. 0
14. Both are 2π
15. 0
16. 1/2
17. 7/6
18. $3(-e^5 + 26)$
19. $\frac{9(e^5+9)}{2}$
20. -1
21. $-1/3$
22. $-5/12$
23. $-1/3$
24. 0
25. 22/15